Practice Problems
Calculus Physics II

Name: Key **Date:** _____

Chapter 52 – RCL Circuits

1) A simple circuit consists only of a 1.0 μF capacitor and a 15 mH coil in series. At what frequency will this circuit oscillate?

$$f = \frac{1}{2\pi\sqrt{LC}} = 1,300\ Hz$$

2) Determine the inductance is needed in series with a 4.7 μF capacitor so the circuit will oscillate at a frequency of 10 kHz?

$$f = \frac{1}{2\pi\sqrt{LC}} \quad \text{Solve for } L \quad L = \frac{1}{(2\pi f)^2 C} = 5.4\times10^{-5}\ H$$

3) What capacitance is needed in series with a 3.7 mH inductor so the circuit will oscillate at a frequency of 1000 Hz?

$$f = \frac{1}{2\pi\sqrt{LC}} \quad C = \frac{1}{(2\pi f)^2 L} = 6.8\times10^{-6}\ F$$

4) A resistor is connected to an ac power supply. In this circuit, the current

A) leads the voltage by 90°.
B) lags the voltage by 90°.
C) lags the voltage by 180°.
(D) is in phase with the voltage.
E) None of the given answers are correct.

5) A simple ac circuit is composed of a resistor connected across the terminals of an ac power source. If the frequency of the source is doubled, what happens to the resistance of the resistor?

A) It increases by a factor of 4.
B) It increases by a factor of 2.
C) It increases by a factor of $\sqrt{2}$
(D) Does not change.
E) It decreases by a factor of 2.
F) It decreases by a factor of 4.

6) A simple ac circuit is composed of a capacitor connected across the terminals of an ac power source. If the frequency of the source is doubled, what happens to the reactance of the capacitor?

A) It increases by a factor of 4.
B) It increases by a factor of 2.
C) It increases by a factor of $\sqrt{2}$
(D) It decreases by a factor of 2.
E) It decreases by a factor of 4.

7) A pure capacitor is connected to an ac power supply. In this circuit, the current

(A) leads the voltage by 90°.
B) lags the voltage by 90°.
C) lags the voltage by 180°.
D) is in phase with the voltage.
E) None of the given answers are correct.

ICE

8) A pure inductor is connected to an ac power supply. In this circuit, the current

A) is in phase with the voltage.
B) leads the voltage by 90°.
C) lags the voltage by 45°.
(D) lags the voltage by 90°.
E) leads the voltage by 45°.

ELI

9) A simple ac circuit consists of an inductor connected across the terminals of an ac power source. If the frequency of the ac source is decreased by a factor of four, what happens to the reactance of the inductor?

(A) It decreases by a factor of four.
B) It decreases by a factor of eight.
C) It increases by a factor of two.
D) It increases by a factor of four.
E) It increases by a factor of eight.

10) As the frequency of the ac voltage across an inductor approaches zero, the inductive reactance of that coil

A) approaches one.
(B) approaches zero.
C) approaches infinity.
D) becomes negative.

11) The reactance of a capacitor is 4.0 kΩ at a frequency of 0.60 kHz. What is the capacitance?

$X_C = \frac{1}{2\pi f C}$ | $C = \frac{1}{2\pi f X_C} = 6.6 \times 10^{-8}\ F$

12) A 6.0 μF capacitor is connected to an ac signal with a frequency of 60 Hz. If the maximum voltage applied to the capacitor is 8.0 V, what is its capacitive reactance?

$X_C = \frac{1}{2\pi f C} = 442\ \Omega$

13) The capacitive reactance of a 44 μF capacitor in an ac circuit is 600 Ω. What is the frequency of the applied signal?

$X_C = \frac{1}{2\pi f C}$ | $f = \frac{1}{2\pi C X_C} = 6\ Hz$

14) What is the reactance of a 20 mH inductor at a frequency of 60 Hz?

$X_L = 2\pi f L = 7.5\ \Omega$

15) At what frequency is the reactance of a 20.0 mH inductor equal to 120 Ω?

$X_L = 2\pi f L \quad | \quad f = \frac{X_L}{2\pi L} = 955\ Hz$

16) What is the inductive reactance of a 2.50 mH coil at 1000 Hz?

$X_L = 2\pi f L = 15.7\ \Omega$

600

17) A 120 V rms signal at ~~60~~ Hz is applied across a series combination of a 30 mH inductor and a 100-Ω resistor. What is the rms value of the current in this circuit?

$X_L = 2\pi f L = 113\ \Omega \quad | \quad Z = \sqrt{R^2 + X_L^2} = 151\ \Omega$

$R = 100\ \Omega \quad | \quad I = \frac{V_{rms}}{Z} = 0.79\ A$

$Z = \sqrt{R^2 + (X_L - X_C)^2}$

18) A 120 V rms signal at 60 Hz is applied across a series combination of a 30 mH inductor and a 100 Ω resistor. What is the rms value of the voltage across the resistor?

$V_R = IR = (0.79A)(100\Omega) = 79\ V$

19) What is the resonance frequency of a series ac circuit consisting of a 40.0 μF capacitor, a 55 Ω resistor, and a 0.030 H inductor?

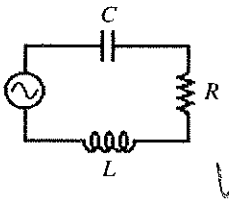
$f = \frac{1}{2\pi\sqrt{LC}} = 145.7\ Hz$

20) In the circuit shown in the figure, the 60 Hz ac source has a voltage amplitude of 120 V, the capacitive reactance is 800 Ω the inductive reactance is 300 Ω and the resistance is 500 Ω.

X_C X_L R

A) What is the capacitance C of the capacitor?

$X_C = \frac{1}{2\pi f C} \quad | \quad C = \frac{1}{2\pi f X_C} = 3.3 \times 10^{-6}\ F$



B) Determine the current through the circuit.

$Z = \sqrt{R^2 + (X_L - X_C)^2} = 707\ \Omega \quad | \quad V = IZ \quad | \quad I = \frac{V}{Z} = 0.17A$

C) Determine the voltage drop across the ~~inductor~~ R, L, or C

$V_L\ V_R = IR = 85\ V$

21) A series *RLC* circuit has a 100 Ω resistor, a 0.1 μF capacitor and a 2.0 mH inductor connected across a 120 V rms ac power source.

A) Determine the voltage across the resistor when the ac power supply's frequency is tuned to 1,000 Hz.

$X_L = 2\pi f L = 12.6\ \Omega$

$X_C = \frac{1}{2\pi f C} = 159\ \Omega$

$R = 100\ \Omega$

$Z = \sqrt{R^2 + (X_L - X_C)^2} = 177.3\ \Omega$

$I = \frac{V}{Z} = 0.677\ A$

$V_R = IR = 67.7\ V$

B) Determine the voltage across the resistor when the ac power supply's frequency is tuned to the resonate frequency.

$f = \frac{1}{2\pi\sqrt{LC}} = 3,559\ Hz$

$X_L = 2\pi f L = 45\ \Omega$

$X_C = \frac{1}{2\pi f C} = 45\ \Omega$

Z at resonance $Z = R$

$I = \frac{V}{R} = 1.2\ A$

$V_R = IR = 120\ V$

22) A series ac circuit has a resonance frequency of 9.0 kHz. If the inductor in the circuit has a value of 2.0 H, and the resistance is 75 Ω, what is the capacitance of the circuit?

$f = \frac{1}{2\pi\sqrt{LC}}$ Solve for C

$\sqrt{C} = \frac{1}{2\pi f \sqrt{L}}$

$C = \left(\frac{1}{2\pi f}\right)^2\left(\frac{1}{L}\right) = \frac{1}{(2\pi f)^2 L} = 1.56 \times 10^{-10}\ F$

KEY

1) 1,300 Hz
2) 5.4×10^{-5} H
3) 6.8×10^{-6} F
4) D
5) D
6) D
7) A
8) D
9) A
10) AB
11) 6.6×10^{-8} F
12) 442 Ω
13) 6 Hz
14) 7.5 Ω
15) 955 Hz
16) 15.7 Ω
17) 0.79 A
18) 79 V
19) 145 Hz
20) A) 3.3×10^{-6} F B) 0.17 A C) 85 V
21) A) 67.7 V B) 120 V
22) $1{,}56 \times 10^{-10}$ F